Cabinet schematic — Pump cabinet

Pump cabinet

Two direct-on-line motor starters, controlled and monitored by a WAGO 750 PLC through two interface relays.

Revision	Date	Description	Created	Checked	Approved
1.1	2026-09-01	First release	SK		
1.2	2026-09-23	Overload feedback to the PLC	SK		
1.3	2026-10-02	TeSys D parts, 24V and GND rails, IEC wire colours	SK		
1.4	2026-10-05	Unit tags, links, busbars and rail bonds, energy on fed supplies	SK		
1.5	2026-10-06	Three phases on four terminals each, pump n on terminal n+2	SK		
1.6	2026-10-06	Owner ruling 2026-10-06, a cabinet is a unit: the place C1 is gone. The change list reads No changes. because the listing drops a place that holds the whole unit, baseline L2, kept by design (REL-DIFF-PLACE)	SK		
1.7	2026-10-16	Re-cut under fransys 0.19.9, drawings redrawn, no design change	SK		
1.8	2026-10-25	Pump contact blocks LADN22: the running feedback leaves on the block's second make contact, 83-84 (R780)	SK		
1.9	2026-10-26	Terminal strip X01 is X4 and breaker F01 is F1: a tag takes no leading zero (R1114)	SK		
1.10	2026-10-28	No mate, board unit or harness line: the interface relays are wired in the cabinet, and the Ethernet link is a cable (R1216, R1253)	SK		
1.11	2026-11-06	Re-release on Fransys v0.22.6; the design is unchanged (R1807).	SK		

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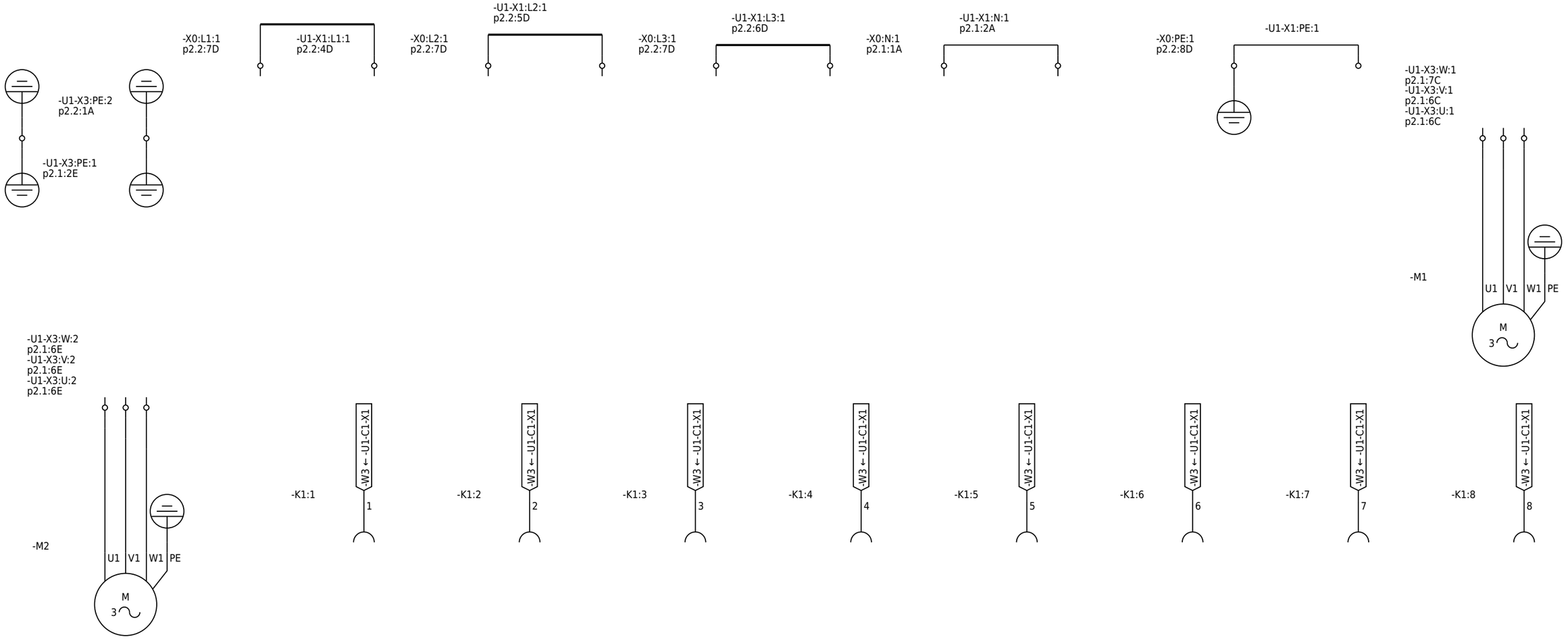
Notes

The upstream board protects the incoming cable with a C16 breaker. The cable lands there on terminals +EXT-X0, the only part of the upstream board in this model.

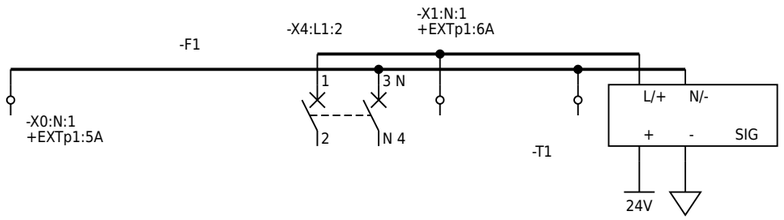
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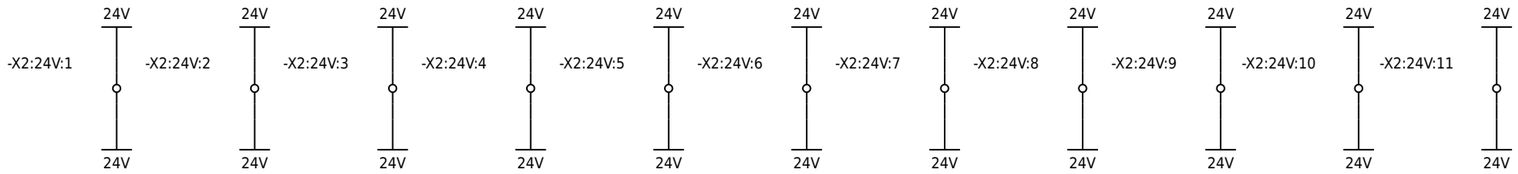
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2.1	Control supply, Pump 1, Pump 2, Incoming supply	5
2.2	Pump 1, Pump 2, WAGO PLC, Incoming supply	6

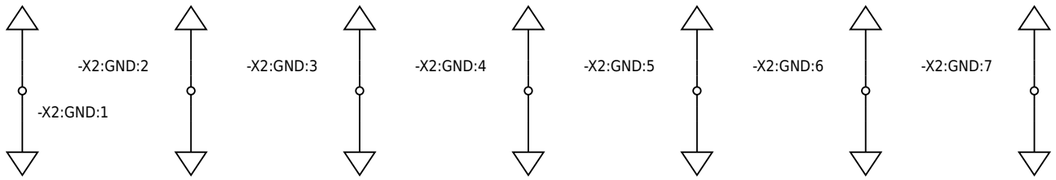
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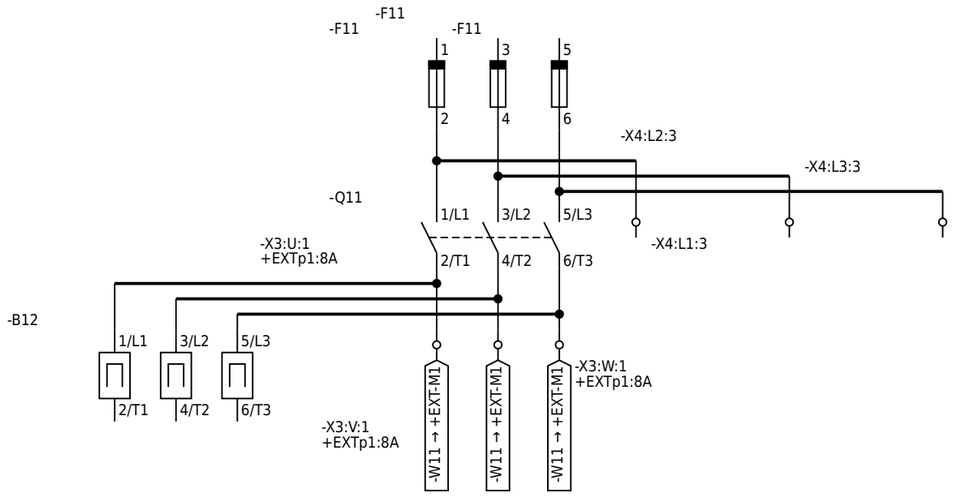


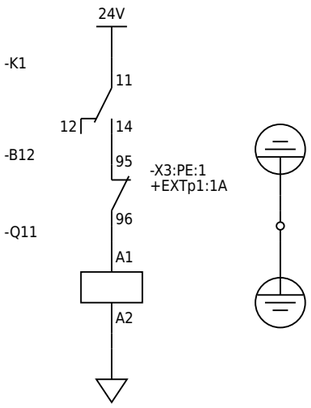
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Customer	Author: SK	Page title: Pump 1, Pump 2, Incoming supply, WAGO PLC	Scope: Pump cabinet	Sheet: 1 / 1	Page: 4 of 10	

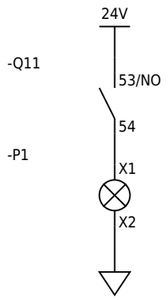


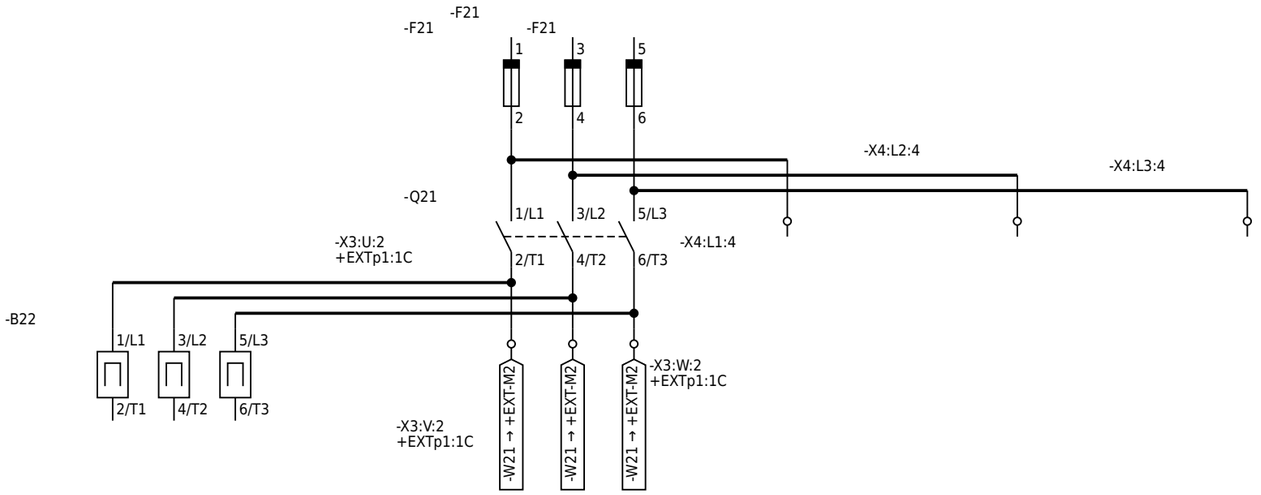




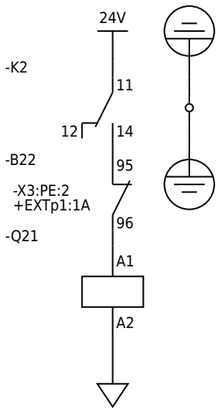


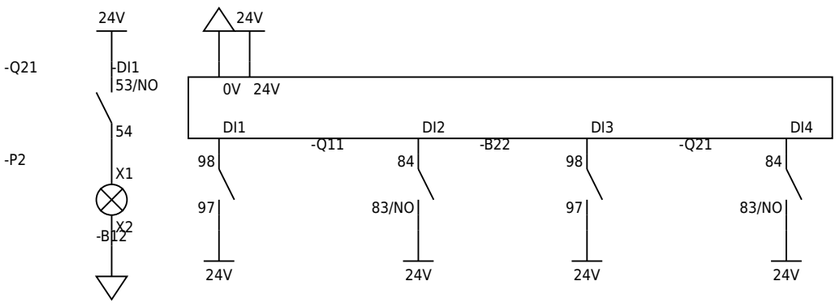


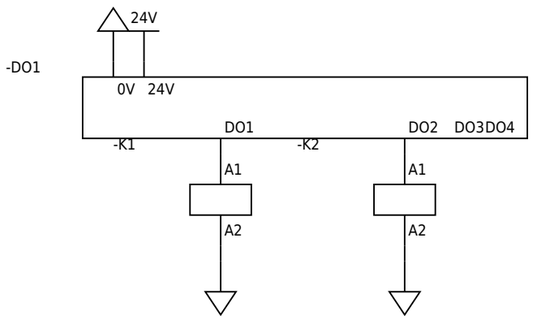


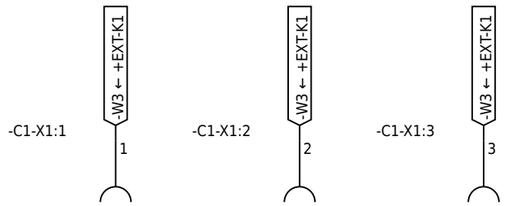


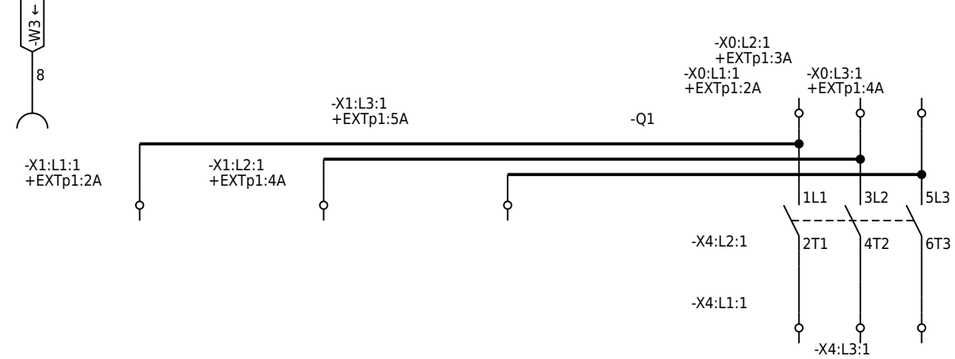
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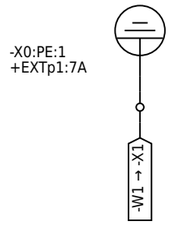


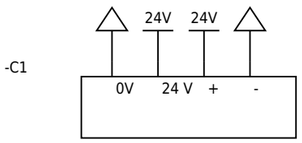












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-DI1

Pin	Channel name	Signal	Signal name	Wired to
-DI1:DI1	1	DI	P1_OVERLOAD	-B12:98
-DI1:DI2	2	DI	P1_RUNNING	-Q11:84
-DI1:DI3	3	DI	P2_OVERLOAD	-B22:98
-DI1:DI4	4	DI	P2_RUNNING	-Q21:84

-DO1

Pin	Channel name	Signal	Signal name	Wired to
-DO1:DO1	1	DO	P1_RUN	-K1:A1
-DO1:DO2	2	DO	P2_RUN	-K2:A1
-DO1:DO3	3	DO	spare	
-DO1:DO4	4	DO	spare	

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-X1

Designation	Group	Index	Side A	Side B
-X1:L1:1	L1	1	-Q1:1L1	+EXT-X0:L1:1
-X1:L2:1	L2	1	-Q1:3L2	+EXT-X0:L2:1
-X1:L3:1	L3	1	-Q1:5L3	+EXT-X0:L3:1
-X1:N:1	N	1	-F1:3 N	+EXT-X0:N:1
-X1:PE:1	PE	1		+EXT-X0:PE:1

-X2

Designation	Group	Index	Side A	Bridge
-X2:24V:1	24V	1	-T1:+	
-X2:24V:2	24V	2	-C1:24 V	
-X2:24V:3	24V	3	-C1:+	
-X2:24V:4	24V	4	-K1:11	
-X2:24V:5	24V	5	-K2:11	
-X2:24V:6	24V	6	-Q11:53/NO	
-X2:24V:7	24V	7	-B12:97	
-X2:24V:8	24V	8	-Q11:83/NO	
-X2:24V:9	24V	9	-Q21:53/NO	
-X2:24V:10	24V	10	-B22:97	
-X2:24V:11	24V	11	-Q21:83/NO	
-X2:GND:1	GND	1	-T1:-	
-X2:GND:2	GND	2	-C1:0V	
-X2:GND:3	GND	3	-C1:-	
-X2:GND:4	GND	4	-Q11:A2	
-X2:GND:5	GND	5	-P1:X2	
-X2:GND:6	GND	6	-Q21:A2	
-X2:GND:7	GND	7	-P2:X2	

-X3

Designation	Group	Index	Side A	Side B
-X3:PE:1	PE	1		+EXT-M1:PE
-X3:PE:2	PE	2		+EXT-M2:PE
-X3:U:1	U	1	-B12:2/T1	+EXT-M1:U1
-X3:U:2	U	2	-B22:2/T1	+EXT-M2:U1
-X3:V:1	V	1	-B12:4/T2	+EXT-M1:V1
-X3:V:2	V	2	-B22:4/T2	+EXT-M2:V1
-X3:W:1	W	1	-B12:6/T3	+EXT-M1:W1
-X3:W:2	W	2	-B22:6/T3	+EXT-M2:W1

-X4

Designation	Group	Index	Side A	Bridge
-X4:L1:1	L1	1	-Q1:2T1	
-X4:L1:2	L1	2	-F1:1	
-X4:L1:3	L1	3	-F11:1	
-X4:L1:4	L1	4	-F21:1	

-X4 (cont.)

Designation	Group	Index	Side A	Bridge
-X4:L2:1	L2	1	-Q1:4T2	
-X4:L2:2	L2	2		
-X4:L2:3	L2	3	-F11:3	
-X4:L2:4	L2	4	-F21:3	
-X4:L3:1	L3	1	-Q1:6T3	
-X4:L3:2	L3	2		
-X4:L3:3	L3	3	-F11:5	
-X4:L3:4	L3	4	-F21:5	

-C1-X1

PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485

Pin	Ends
1	+EXT-K1:1
2	+EXT-K1:2
3	+EXT-K1:3
4	+EXT-K1:4
5	+EXT-K1:5
6	+EXT-K1:6
7	+EXT-K1:7
8	+EXT-K1:8

PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485

-C1-X2: no pin wired

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Mpn	Manufacturer	Description	Count	Designations
2002-1201	WAGO	TOPJOB S feed-through terminal block, 2.5 mm2, grey; 24 A; 800 V	40	-X1:L1:1; -X1:L2:1; -X1:L3:1; -X1:N:1; -X2:24V:1; -X2:24V:2; -X2:24V:3; -X2:24V:4; -X2:24V:5; -X2:24V:6; -X2:24V:7; -X2:24V:8; -X2:24V:9; -X2:24V:10; -X2:24V:11; -X2:GND:1; -X2:GND:2; -X2:GND:3; -X2:GND:4; -X2:GND:5; -X2:GND:6; -X2:GND:7; -X3:U:1; -X3:U:2; -X3:V:1; -X3:V:2; -X3:W:1; -X3:W:2; -X4:L1:1; -X4:L1:2; -X4:L1:3; -X4:L1:4; -X4:L2:1; -X4:L2:2; -X4:L2:3; -X4:L2:4; -X4:L3:1; -X4:L3:2; -X4:L3:3; -X4:L3:4
2002-1207	WAGO	TOPJOB S ground terminal block, 2.5 mm2, green-yellow	3	-X1:PE:1; -X3:PE:1; -X3:PE:2
3LD2054-0TK51	Siemens	Switch-disconnector, 3-pole; 16 A	1	-Q1
3NW7033	Siemens	Fuse holder, 3-pole, 10x38 mm; 32 A	2	-F11; -F21
3NW8004-1	Siemens	SENTRON cylindrical fuse link, 10x38 mm, aM; 4 A	6	-F11; -F21
3SU1102-6AA40-1AA0	Siemens	Pilot light 22 mm, green, integrated LED 24 V AC/DC	2	-P1; -P2
40.61.9.024.1000	Finder	PCB relay, 1 changeover contact, 16 A, 24 V DC coil	2	-K1; -K2
5SY6506-7	Siemens	Miniature circuit breaker, 1P+N, curve C; 6 A	1	-F1
750-402	WAGO	4-channel digital input module, 24 V DC	1	-DI1
750-504	WAGO	4-channel digital output module, 24 V DC; 0.5 A	1	-DO1
750-600	WAGO	End module, terminates the internal data bus; no data contacts	1	-E1
750-8212	WAGO	PFC200 controller, 2nd generation, 2x ETHERNET, RS-232/-485	1	-C1
LADN22	Schneider Electric	TeSys D front-mount auxiliary contact block, 2 NO + 2 NC	2	-Q11; -Q21
LC1D09BD	Schneider Electric	TeSys D contactor, 3-pole, AC-3 9 A, 1 NO + 1 NC auxiliary, 24 V DC coil	2	-Q11; -Q21
LRD08	Schneider Electric	TeSys thermal overload relay, class 10A, 2.5-4 A, 1 NO + 1 NC	2	-B12; -B22
QUINT4-PS/1AC/24DC/2.5/SC	Phoenix Contact	Power supply, 1-phase, 24 V DC; 2.5 A	1	-T1

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